Networking Cables Assignment

Ethernet | Fiber Optic | T568B Wiring | RJ45 Crimping

Name: _____ Class: _____ Date: _____

This report answers all four tasks in simple language. Each section has a short explanation, a diagram or table where useful, and a quick tip to help you remember the key points.

1 Ethernet vs Fiber Optic Cables

Both cables carry network data, but they do it in very different ways. Here are the **3 key differences**:

Difference	Ethernet cable (copper)	Fiber optic cable
1. How data travels	Sends data as **electrical signals** through 8 thin copper wires twisted in 4 pairs (e.g. Cat5e, Cat6).	Sends data as **pulses of light** through a hair-thin strand of glass (or plastic) called the core.
2. Distance and speed	Works up to **100 metres** per run. Cat6 gives 1 Gbps (10 Gbps on very short runs).	Works for **hundreds of metres to many kilometres** and easily reaches 10 Gbps, 40 Gbps or more.
3. Interference, cost and installation	Can pick up electrical noise from motors or power lines. **Cheap** and easy to install with a simple RJ45 crimper.	**Immune** to electrical interference and more secure. Costs more and needs special tools (cleaver, splicer) and skills.

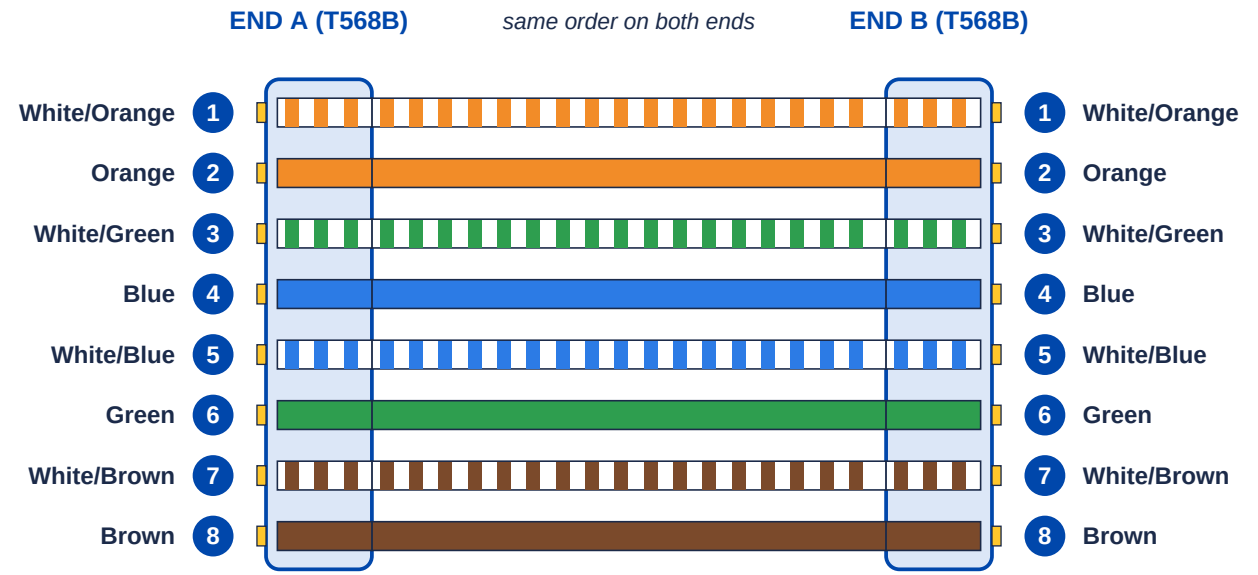
Ethernet example
Home WiFi router to a desktop PC or smart TV. The distance is short, speed needs are modest, and a cheap Cat5e/Cat6 cable does the job perfectly.

Fiber optic example
College campus backbone linking the library, labs and hostels. Buildings are far apart and many users share the link, so fiber's long reach and huge speed are needed.

Easy way to remember: Ethernet = electricity, short distance, low cost. Fiber = light, long distance, high speed.

2 Straight-Through Ethernet Cable: T568B Wiring

A **straight-through** cable has the **same wire order on both ends**. It is used to connect different devices, such as a PC to a switch or a router to a PC. The T568B standard sets this order:



Plug held with the clip facing DOWN and the gold contacts facing YOU: pin 1 is on the LEFT.

Pin	Wire colour	Sample	Pair	Job
1	White/Orange		Orange pair	Transmit data (TX+)
2	Orange		Orange pair	Transmit data (TX-)
3	White/Green		Green pair	Receive data (RX+)
4	Blue		Blue pair	Extra pair (used in Gigabit)
5	White/Blue		Blue pair	Extra pair (used in Gigabit)
6	Green		Green pair	Receive data (RX-)
7	White/Brown		Brown pair	Extra pair (used in Gigabit)
8	Brown		Brown pair	Extra pair (used in Gigabit)

Memory trick (T568B, pins 1 to 8): W-O, O, W-G, B, W-B, G, W-Br, Br. Notice that **blue stays in the middle (pins 4 and 5)**, and green is split around it (pins 3 and 6). That split is normal and is what keeps the signal clean.

3 How to Attach an RJ45 Connector to a CAT6 Cable

Tools and parts: CAT6 cable, RJ45 connectors (CAT6 rated), RJ45 crimping tool, wire cutter/stripper, and a cable tester.

1	**Cut the cable end clean** Use the cutter to trim the end of the cable straight, so the wires are even and undamaged.
2	**Strip the outer jacket** Remove about **2.5 cm (1 inch)** of the outer jacket with the stripper. Rotate gently so you do not nick the thin wires inside.
3	**Remove the plastic divider** CAT6 has a plastic cross (spine) between the pairs. Cut it off flush with the jacket, because it will not fit in the connector.
4	**Untwist and straighten** Separate the 4 pairs and untwist them. Straighten each wire with your fingers so it lies flat.
5	**Arrange in T568B order** Line up the wires left to right: **White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown, Brown**. Hold them tightly together.
6	**Trim the wires level** Cut all 8 wires evenly so that about **1.2 cm (half an inch)** sticks out from the jacket.
7	**Insert into the connector** Hold the plug with the clip facing down and slide the wires in. Push firmly until each wire touches the very front of the plug. The jacket should go slightly inside the plug.
8	**Check before you crimp** Look at the end of the plug: you should see 8 copper tips at the front, and the colours should still be in order. If not, pull out and redo.
9	**Crimp it** Place the plug into the RJ45 slot of the crimper and squeeze hard until it fully closes. This presses the gold pins into the wires and locks the jacket.
10	**Do the other end and test** Repeat steps 1 to 9 with the same T568B order. Then plug both ends into a cable tester: lights 1 to 8 should glow in order.

Common mistakes: stripping too much jacket (cable becomes unreliable), wires not reaching the front of the plug (no contact), and mixing up the colour order (cable fails the tester). If the tester shows a problem, cut off the plug and start again.

Video watched: _____ (add the title or link of the YouTube tutorial you watched).

4 Gaming Cafe with 10 PCs: My Choice

Cable chosen
Ethernet (Cat6)

Tool chosen
RJ45 crimper

My explanation

For a gaming cafe with 10 PCs in one room, I would use **Cat6 Ethernet cables** with an **RJ45 crimper**. Every PC will be only a few metres from the switch, well inside the 100 m limit, so fiber's long-distance advantage would be wasted. Cat6 gives a fast and stable 1 Gbps connection with very low delay (ping), which is exactly what LAN games need. It is also much cheaper: Ethernet cables, a crimper and a gigabit switch cost far less than fiber cables, a cleaver, and fiber-ready network cards, and the cafe owner can easily fix or replace a cable on the spot.

Factor	Ethernet (Cat6)	Fiber optic	Better for the cafe
Speed	1 Gbps, enough for gaming	10 Gbps or more, not needed	Ethernet (enough)
Distance	Up to 100 m	Kilometres	Ethernet (rooms are small)
Budget	Low cost, cheap tools	High cost, special tools	Ethernet

Conclusion: Choose the tool that fits the job. Ethernet is the smart, budget-friendly choice for a small LAN, while fiber is best kept for long, high-capacity links such as a campus backbone or data center.